

Problem 1.0: Schrödinger Equation in Different Basis



As a warm-up question, starting from the time-dependent Schrödinger equation in operator form,

$$i\hbar \frac{d}{dt} |\psi(t)\rangle = \hat{H} |\psi(t)\rangle,$$

where $\hat{H} = \frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}})$, we will derive the “standard” Schrödinger equation in terms of the position-space and momentum-space wavefunctions. Conceptually, this is a trivial exercise, but it does require some elbow grease, such as much of what is to come in this course.

- Derive the time-dependent Schrödinger equation in position space (in 3 spatial dimensions), i.e., find the differential equation obeyed by the wavefunction $\psi(\mathbf{x}, t) = \langle \mathbf{x} | \psi(t) \rangle$. (Hint: use the completeness relation of the position basis, $\int d^3\mathbf{x} |\mathbf{x}\rangle \langle \mathbf{x}| = \hat{I}$, and the fact that the momentum operator in position space is $\hat{\mathbf{p}} = -i\hbar \nabla$. You can assume that the potential operator depends exclusively on powers of $\hat{\mathbf{x}}$, such that $V(\hat{\mathbf{x}}) |\mathbf{x}\rangle = V(\mathbf{x}) |\mathbf{x}\rangle$.)
- Repeat the same steps to derive the time-dependent Schrödinger equation in momentum space, i.e., find the *integro-differential* equation obeyed by $\phi(\mathbf{p}, t) = \langle \mathbf{p} | \psi(t) \rangle$. (Hint: recall that the momentum and position kets are related by the Fourier transform, $\langle \mathbf{x} | \mathbf{p} \rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}$.)
- Now, using the Fourier transform relations between position and momentum space wavefunctions,

$$\psi(\mathbf{x}, t) = \frac{1}{(2\pi\hbar)^{3/2}} \int d^3\mathbf{p} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar} \phi(\mathbf{p}, t),$$

$$\phi(\mathbf{p}, t) = \frac{1}{(2\pi\hbar)^{3/2}} \int d^3\mathbf{x} e^{-i\mathbf{p}\cdot\mathbf{x}/\hbar} \psi(\mathbf{x}, t),$$

show that the two Schrödinger equations you derived in (a) and (b) are equivalent.

(Hint: start from the position space Schrödinger equation. Recall that the delta function can be written as $\delta(\mathbf{p} - \mathbf{p}') = \frac{\hbar^3}{(2\pi)^3} \int d^3\mathbf{x} e^{i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}/\hbar}$.)



Solution to Problem 1.0: This problem is all about letting the formalism do the work and gives us a glimpse of how powerful the basis-independent Schrödinger equation is.

a) We want to express everything in terms of the position-space wavefunction $\psi(\mathbf{x}, t) = \langle \mathbf{x} | \psi(t) \rangle$. To that end, we will contract both sides of the operator Schrödinger equation with $\langle \mathbf{x} |$ from the left (i.e., apply the linear operator $\langle \mathbf{x} |$ to both sides):¹

$$i\hbar \frac{\partial}{\partial t} \langle \mathbf{x} | \psi(t) \rangle = \langle \mathbf{x} | \hat{H} | \psi(t) \rangle \quad (1)$$

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{x}, t) = \langle \mathbf{x} | \left(\frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}}) \right) | \psi(t) \rangle. \quad (2)$$

where we just used the definition of the position-space wavefunction.

The RHS is more interesting. Let us insert the completeness relation of the position basis right before the ket $|\psi(t)\rangle$:

$$\int d^3\mathbf{x}' \langle \mathbf{x} | \left(\frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}}) \right) | \mathbf{x}' \rangle \langle \mathbf{x}' | \psi(t) \rangle = \int d^3\mathbf{x}' \langle \mathbf{x} | \left(\frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}}) \right) | \mathbf{x}' \rangle \psi(\mathbf{x}', t).$$

Curiously, our RHS became a sort of expectation value times the wavefunction. Let us find what this expectation value is. First, consider the potential term. This one is assumed to be simple enough that $V(\hat{\mathbf{x}}) | \mathbf{x}' \rangle = V(\mathbf{x}') | \mathbf{x}' \rangle$. Now, moving onto the kinetic term, we can use the fact that in position space, the momentum operator is $\hat{\mathbf{p}} = -i\hbar \nabla$, which ought to return a 3-D vector. Thus, $\hat{\mathbf{p}}^2 = -\hbar^2 \nabla^2$, which ought to return a scalar. Putting this all together, we have

$$\langle \mathbf{x} | \left(\frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}}) \right) | \mathbf{x}' \rangle = \langle \mathbf{x} | \left(-\frac{\hbar^2}{2m} \nabla'^2 + V(\mathbf{x}') \right) | \mathbf{x}' \rangle \quad (3)$$

$$= \left(-\frac{\hbar^2}{2m} \nabla'^2 + V(\mathbf{x}') \right) \langle \mathbf{x} | \mathbf{x}' \rangle \quad (4)$$

$$= \left(-\frac{\hbar^2}{2m} \nabla'^2 + V(\mathbf{x}') \right) \delta(\mathbf{x} - \mathbf{x}'), \quad (5)$$

where we used the fact that $\langle \mathbf{x} | \mathbf{x}' \rangle = \delta(\mathbf{x} - \mathbf{x}')$. Under the integral, we can now perform the integration over $\mathbf{x}'$ using the properties of the Dirac delta to find

$$\langle \mathbf{x} | \hat{H} | \psi(t) \rangle = \int d^3\mathbf{x}' \left(-\frac{\hbar^2}{2m} \nabla'^2 + V(\mathbf{x}') \right) \delta(\mathbf{x} - \mathbf{x}') \psi(\mathbf{x}', t) = \left(-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{x}) \right) \psi(\mathbf{x}, t). \quad (6)$$

The desired result follows:

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{x}, t) = \left(-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{x}) \right) \psi(\mathbf{x}, t).$$

b) We will repeat the same steps, but now using the momentum basis. Contracting both sides of the operator Schrödinger equation with $\langle \mathbf{p} |$ from the left, we have

$$i\hbar \frac{\partial}{\partial t} \langle \mathbf{p} | \psi(t) \rangle = \langle \mathbf{p} | \hat{H} | \psi(t) \rangle \quad (7)$$

$$i\hbar \frac{\partial}{\partial t} \phi(\mathbf{p}, t) = \langle \mathbf{p} | \left(\frac{\hat{\mathbf{p}}^2}{2m} + V(\hat{\mathbf{x}}) \right) | \psi(t) \rangle. \quad (8)$$

¹Note that what we mean by the position kets (and therefore the bras) is definitely time-independent, so we can place the bra inside the time derivative, which now becomes a *partial* derivative since the wavefunction depends now on both time and space.

Let us insert the completeness relation of the momentum basis right before the ket $|\psi(t)\rangle$ to get the RHS of Eq. (8):

$$\text{RHS} = \int d^3\mathbf{p}' \langle \mathbf{p} | \left(\frac{\hat{p}^2}{2m} + V(\hat{\mathbf{x}}) \right) | \mathbf{p}' \rangle \langle \mathbf{p}' | \psi(t) \rangle = \int d^3\mathbf{p}' \langle \mathbf{p} | \left(\frac{\hat{p}^2}{2m} + V(\hat{\mathbf{x}}) \right) | \mathbf{p}' \rangle \phi(\mathbf{p}', t).$$

Now, it is the kinetic term that becomes trivial, since $\hat{\mathbf{p}} |\mathbf{p}'\rangle = \mathbf{p}' |\mathbf{p}'\rangle$. The potential term, on the other hand, is trickier now since we do not know what form it takes in momentum space. This is not particularly surprising since the whole point of working in momentum space should be that we know how to write down the problem in momentum space, but we are given a potential in position space. Let us fix this by inserting the completeness relation of the position basis right after the potential operator:

$$\langle \mathbf{p} | V(\hat{\mathbf{x}}) | \mathbf{p}' \rangle = \int d^3\mathbf{x}' \langle \mathbf{p} | V(\hat{\mathbf{x}}) | \mathbf{x}' \rangle \langle \mathbf{x}' | \mathbf{p}' \rangle \quad (9)$$

$$= \int d^3\mathbf{x}' V(\mathbf{x}') \langle \mathbf{p} | \mathbf{x}' \rangle \langle \mathbf{x}' | \mathbf{p}' \rangle. \quad (10)$$

If you recall, the momentum and position kets are related by the Fourier transform,²

$$\langle \mathbf{x} | \mathbf{p} \rangle = \frac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}.$$

Therefore, we have that the sandwich above is just the Fourier transform of the potential(!):

$$\langle \mathbf{p} | V(\hat{\mathbf{x}}) | \mathbf{p}' \rangle = \int d^3\mathbf{x}'' V(\mathbf{x}'') \frac{1}{(2\pi\hbar)^{3/2}} e^{-i\mathbf{p}\cdot\mathbf{x}''/\hbar} \frac{1}{(2\pi\hbar)^{3/2}} e^{i\mathbf{p}'\cdot\mathbf{x}''/\hbar} \quad (11)$$

$$= \frac{1}{(2\pi\hbar)^3} \int d^3\mathbf{x}'' V(\mathbf{x}'') e^{i(\mathbf{p}'-\mathbf{p})\cdot\mathbf{x}''/\hbar} \equiv \tilde{V}(\mathbf{p}' - \mathbf{p}), \quad (12)$$

where we made it apparent that $\tilde{V}(\mathbf{q})$ is simply the Fourier transform of the potential $V(\mathbf{x})$. Okay, now we are really working in momentum space!

Putting this all together, we have

$$\langle \mathbf{p} | \hat{H} | \psi(t) \rangle = \int d^3\mathbf{p}' \left(\frac{\mathbf{p}'^2}{2m} + \tilde{V}(\mathbf{p}' - \mathbf{p}) \right) \phi(\mathbf{p}', t). \quad (13)$$

The desired result follows:

$$i\hbar \frac{\partial}{\partial t} \phi(\mathbf{p}, t) = \frac{\mathbf{p}^2}{2m} \phi(\mathbf{p}, t) + \int d^3\mathbf{p}' \tilde{V}(\mathbf{p}' - \mathbf{p}) \phi(\mathbf{p}', t).$$

c) To show that the two Schrödinger equations are equivalent, we will start from the position-space Schrödinger equation and use the Fourier transform relations to get to the momentum-space Schrödinger equation.

Starting from the position-space Schrödinger equation,

$$i\hbar \frac{\partial}{\partial t} \psi(\mathbf{x}, t) = \left(-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{x}) \right) \psi(\mathbf{x}, t),$$

we will substitute the Fourier transform relation for $\psi(\mathbf{x}, t)$:

$$i\hbar \frac{\partial}{\partial t} \left(\frac{1}{(2\pi\hbar)^{3/2}} \int d^3\mathbf{p} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar} \phi(\mathbf{p}, t) \right) = \left(-\frac{\hbar^2}{2m} \nabla^2 + V(\mathbf{x}) \right) \left(\frac{1}{(2\pi\hbar)^{3/2}} \int d^3\mathbf{p} e^{i\mathbf{p}\cdot\mathbf{x}/\hbar} \phi(\mathbf{p}, t) \right).$$

²See Sec. 1.7.3 in Sakurai and Napolitano. The normalization constant is fixed by the normalization condition $\langle \mathbf{p} | \mathbf{p}' \rangle = \delta(\mathbf{p} - \mathbf{p}')$.

Next, we will multiply both sides by $\frac{1}{(2\pi\hbar)^{3/2}}e^{-i\mathbf{p}'\cdot\mathbf{x}/\hbar}$ and integrate over all space $d^3\mathbf{x}$:

$$i\hbar\frac{\partial}{\partial t}\int d^3\mathbf{x}\frac{1}{(2\pi\hbar)^3}\int d^3\mathbf{p}e^{i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}/\hbar}\phi(\mathbf{p},t) \quad (14)$$

$$= \int d^3\mathbf{x}\frac{e^{-i\mathbf{p}'\cdot\mathbf{x}/\hbar}}{(2\pi\hbar)^3}\left(-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{x})\right)\int d^3\mathbf{p}e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}\phi(\mathbf{p},t). \quad (15)$$

The LHS is straightforward, since the integral over $\mathbf{x}$ just gives us a Dirac delta:

$$\text{LHS} = i\hbar\frac{\partial}{\partial t}\int d^3\mathbf{p}\delta(\mathbf{p}-\mathbf{p}')\phi(\mathbf{p},t) = i\hbar\frac{\partial}{\partial t}\phi(\mathbf{p}',t). \quad (16)$$

The RHS is more involved. Let us split it into the kinetic and potential terms. The kinetic term is (noting that ∇ acts on $\mathbf{x}$ and the exponential we inserted on both sides is not acted upon by the derivative):

$$\text{RHS}_{\text{kin}} = -\frac{\hbar^2}{2m}\int d^3\mathbf{x}\frac{e^{-i\mathbf{p}'\cdot\mathbf{x}/\hbar}}{(2\pi\hbar)^3}\nabla^2\int d^3\mathbf{p}e^{i\mathbf{p}\cdot\mathbf{x}/\hbar}\phi(\mathbf{p},t) \quad (17)$$

$$= -\frac{\hbar^2}{2m}\int d^3\mathbf{p}\phi(\mathbf{p},t)\int d^3\mathbf{x}\frac{e^{-i\mathbf{p}'\cdot\mathbf{x}/\hbar}}{(2\pi\hbar)^3}\nabla^2e^{i\mathbf{p}\cdot\mathbf{x}/\hbar} \quad (18)$$

$$= -\frac{\hbar^2}{2m}\int d^3\mathbf{p}\phi(\mathbf{p},t)\int d^3\mathbf{x}\frac{e^{i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}/\hbar}}{(2\pi\hbar)^3}\left(-\frac{\mathbf{p}^2}{\hbar^2}\right) \quad (19)$$

$$= \int d^3\mathbf{p}\phi(\mathbf{p},t)\frac{\mathbf{p}^2}{2m}\delta(\mathbf{p}-\mathbf{p}') = \frac{\mathbf{p}'^2}{2m}\phi(\mathbf{p}',t). \quad (20)$$

The potential term is

$$\text{RHS}_{\text{pot}} = \int d^3\mathbf{x}\frac{1}{(2\pi\hbar)^3}V(\mathbf{x})\int d^3\mathbf{p}e^{i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}/\hbar}\phi(\mathbf{p},t) \quad (21)$$

$$= \int d^3\mathbf{p}\phi(\mathbf{p},t)\int d^3\mathbf{x}\frac{1}{(2\pi\hbar)^3}V(\mathbf{x})e^{i(\mathbf{p}-\mathbf{p}')\cdot\mathbf{x}/\hbar} \quad (22)$$

$$= \int d^3\mathbf{p}\phi(\mathbf{p},t)\tilde{V}(\mathbf{p}-\mathbf{p}'), \quad (23)$$

where we recognized the Fourier transform of the potential once again. Putting this all together, we have

$$i\hbar\frac{\partial}{\partial t}\phi(\mathbf{p}',t) = \left(\frac{\mathbf{p}'^2}{2m} + \int d^3\mathbf{p}\tilde{V}(\mathbf{p}-\mathbf{p}')\phi(\mathbf{p},t)\right),$$

which is exactly the momentum-space Schrödinger equation we derived in part (b), thus proving the equivalence of the two formulations.